Mask R-CNN GPU Webcam Detection

Commands, Real-Time Shortcuts & Hyperparameters Guide

Hardware / Engine: NVIDIA GeForce RTX 4060 GPU | PyTorch 2.6.0 (CUDA 12.4) | ResNet-50 FPN V2

1. Real-Time Keyboard Shortcuts (Live Webcam HUD)

Press any of the following keys while the OpenCV webcam display window is focused:

Key	Action Name	Description & Effect	Category
M	Toggle Masks	Turn colored instance segmentation masks ON / OFF	Display
B	Toggle Boxes	Turn bounding box rectangle overlays ON / OFF	Display
L	Toggle Labels	Turn class names and confidence percentage pills ON / OFF	Display
+ or =	Increase Conf	Raise confidence threshold by **+0.05** (up to 0.95)	Threshold
- or _	Decrease Conf	Lower confidence threshold by **-0.05** (down to 0.10)	Threshold
[	Decrease Opacity	Make segmentation masks more transparent (-0.05 alpha)	Styling
]	Increase Opacity	Make segmentation masks more opaque (+0.05 alpha)	Styling
S	Snapshot	Save high-res frame with detections to captures/snapshot_*.png	Capture
R	Record Video	Toggle MP4 video recording on/off (pulsing red REC indicator)	Capture
H	Help Modal	Open / close semi-transparent on-screen shortcuts help dialog	HUD
Q / ESC	Quit	Cleanly release camera, stop recording, and exit application	System

2. Launch & Execution Commands

You can launch the project using one-click scripts or terminal commands:

Scenario / Command	Command Syntax	Details
1-Click Windows	`run.bat`	Double-click in Explorer or run from CMD prompt
PowerShell	`.\run.ps1`	Direct execution in Windows PowerShell / pwsh
Default Launch	`.venv\Scripts\python.exe main.py`	Loads all hyperparameters directly from config.yaml
Custom Config File	`python main.py --config custom.yaml`	Use a specific YAML configuration preset
Change Resolution	`python main.py --width 1920 --height 1080`	Overrides resolution to 1080p Full HD
Select Camera	`python main.py --camera 1`	Connects to second / external USB camera index
Process Video File	`python main.py --video sample.mp4`	Runs Mask R-CNN segmentation on a recorded video
Test GPU / CUDA	`.venv\Scripts\python.exe test_gpu.py`	Runs CUDA tensor speed benchmark on RTX 4060

3. Central Hyperparameters Reference (config.yaml)

All runtime behavior is controlled by **config.yaml**. Edit this file to customize without touching any code:

Section	Hyperparameter	Default	Description & Valid Values
model	`version`	"v2"	Architecture version: "v2" (ResNet-50 FPN V2) or "v1"
model	`device`	"cuda"	Hardware target: "cuda" (RTX GPU) or "cpu"
model	`use_fp16`	true	Mixed precision inference for maximum FPS and lower VRAM
detection	`conf_threshold`	0.50	Minimum confidence score to detect an object (0.05 to 0.95)
detection	`mask_threshold`	0.50	Pixel probability cutoff for mask binarization (0.1 to 0.9)
detection	`target_classes`	[]	List of class names to detect (e.g. ["person", "laptop"]). Empty = all
camera	`index`	0	Webcam hardware index (0 = primary, 1 = USB webcam)
camera	`width / height`	1280 / 720	Webcam capture resolution in pixels
visualization	`mask_alpha`	0.45	Opacity of segmentation masks (0.10 transparent to 0.90 solid)
visualization	`box_thickness`	2	Thickness in pixels for bounding box outlines
captures	`save_dir`	"captures"	Folder path where snapshots and video recordings are stored

4. Supported COCO Classes (80 Categories)

People & Vehicles: person, bicycle, car, motorcycle, airplane, bus, train, truck, boat
Outdoor & Signals: traffic light, fire hydrant, stop sign, parking meter, bench
Animals: bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe
Accessories & Travel: backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket
Indoor & Electronics: bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush